Technical Data Sheet: KNSB Propellant Grain Fabrication Log

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Abstract

This technical document details the empirical manufacturing logs, thermal adjustments, and final physical metrics of KNSB propellant Batch 02 processed on Monday, 6th July 2026. This batch achieved an operational saturation point at a dry fuel mass of 591.0 g, requiring a proportional oxidizer integration of 1098.0 g. Following successful casting and cooling protocols, this specific grain has been designated as the formal experimental control baseline to support direct comparative performance tracking in upcoming static test fires.

Section 1: Achieved Batch Specifications Checklist

The real-world metrics, scale restrictions, and physical dimensions captured during the casting run are organized in Table II.

Table II: Achieved Specifications and Physical Metrics

Parameter / Identifier	Target Specification	Achieved Metric
Experimental Role	Static Test Baseline Control	Established Control Grain
Propellant Formulation	KNSB (65:35)	65.01% Oxidizer : 34.99% Fuel
Initial Fuel Mass (Sorbitol 70)	800.0 g	800.0 g
Final Processed Fuel	560.0 g (Target)	591.0 g (Operational Saturation)

Mass		
Measured Oxidizer Mass (KNO_3)	1097.57 g (Calc)	1098.0 g (1g balance resolution)
Slurry Blending Temperature	90°C (30 min)	90°C (Thermal Stability Maintained)
Actual Casting Temperature	60°C	60°C (Poured Into Liner)
Liner Tare Weight	Standard Baseline	145.0 g
Measured Propellant Net Mass	Calculated Yield	1205.0 g (After subtraction)
Total Assembly Gross Mass	Completed System	1305.0 g
Grain Outer Radius	8.60 cm	8.60 cm
Grain Internal Core Radius	3.55 cm	3.55 cm
Grain Column Length	14.50 cm	14.50 cm
Density	1.84g/cm³	1.715 g/cm³